

Data Management
2025/2026

Projects

Instructions and Guidelines

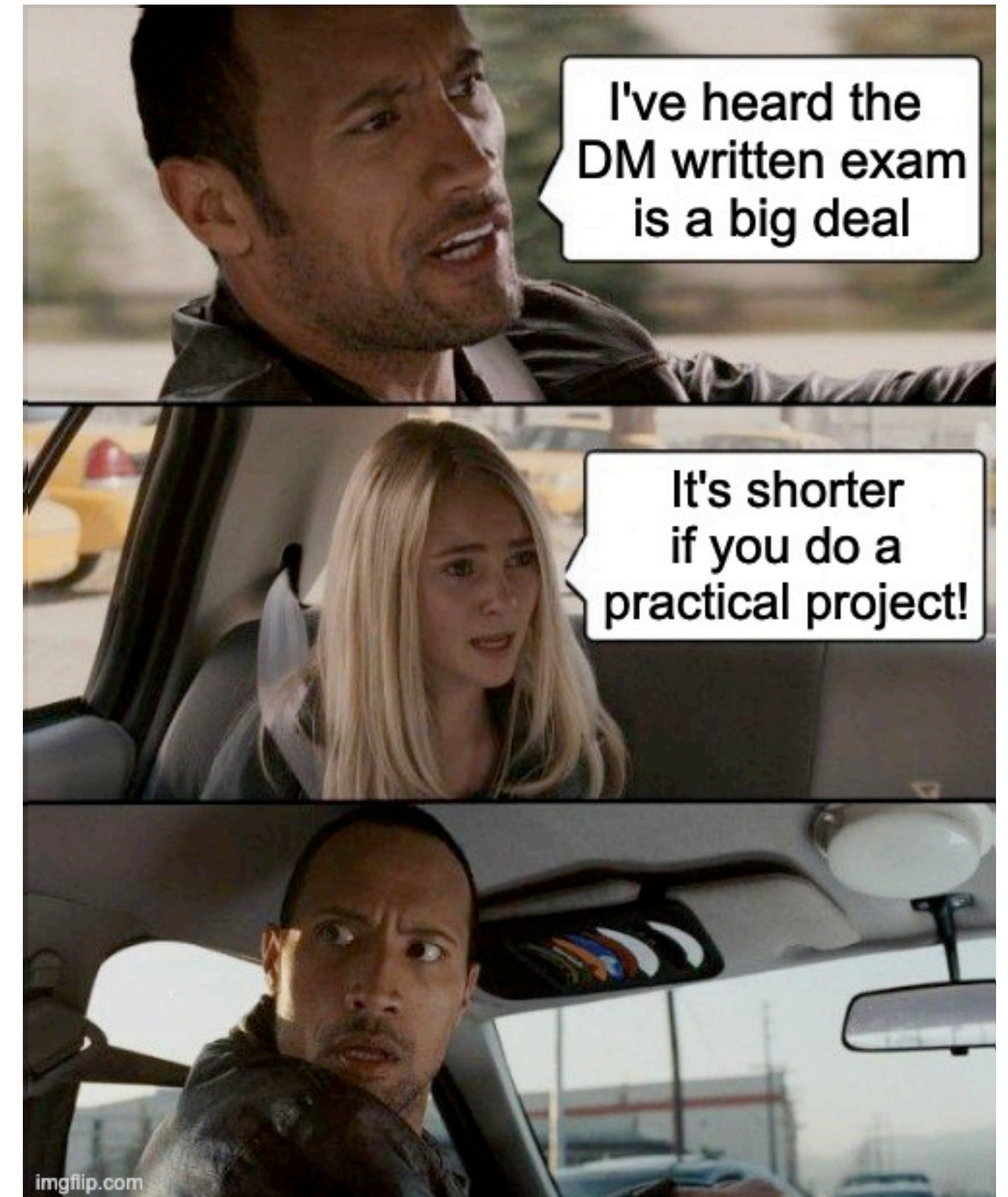


SAPIENZA
UNIVERSITÀ DI ROMA

Tutor
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Topics

- General instructions
- Project approval and presentation modalities
- How to write a project proposal
- How to ask for project presentation
- Evaluation
- Deadlines
- Possible ideas for projects
- Useful resources



Options to pass the exam

Two modalities exist to pass the Data Management exam: written exam only (**Option 1**) or project + written exam (**Option 2**).

Differences between the two options:

1. **Option 1**: full standard written exam
2. **Option 2**: shortened written exam (passing score $x \in [15,24]$) plus a practical project (passing score $y \in [4,8]$). Final grade $z = x + y$ ($z \in [19,32]$)

NB: projects can be carried out either **individually** or in **groups of two students**

More details can be found in the [course website](#)

Procedure for choosing the Exam Modality

- Students willing to do the exam according to **Option 2** (written exam + project) must send a project proposal to delfino@diag.uniroma1.it **BEFORE** booking for the exam.
- If you choose **Option 1** (only written exam), just book for the written exam, without sending me an email to make your choice explicit.

NB: doing the exam without having communicated the choice of selecting **Option 2**, automatically implies that you chose **Option 1**, and you will be evaluated accordingly, without exceptions.

Projects approval and presentation

If you choose **Option 2** (written exam + project), please do the following once ready:

1. Send me your project proposal* (delfino@diag.uniroma1.it) and wait for it to be approved (do not start working before receiving my feedback)
2. If your proposal is accepted, start working on it
3. After completing the project, prepare a 10 to 15 mins slideshow presentation (stick to the time limit!) + a 5 minutes live demo
4. Send me an e-mail containing the material* of your project specifying whether you prefer to have your presentation in person or remotely

* *details in the next slides*

Project proposal

Write a single-page pdf document including:

- Names and student IDs (matricola) of the group members
- Chosen type of project (DW, DBMS comparison, use of a NoSQL tool, etc.)
- Description of the dataset(s) which you intend to use (include a reference link)
- Brief description (5-10 lines) of the work you intend to do

Send me an e-mail having:

- **[DM] Project Proposal** as the subject
- The pdf document described before as an attachment
- All group members participating in the conversation (either as the sender or in cc)

Project presentation

Once you are ready to have present your project, send me an e-mail having:

- **[DM] Project Discussion** as the subject
- Your slides as an attachment
- Any link to additional material you think might be useful (repository, data, etc.)

I will reply by either proposing you an appointment to present your project, or suggesting you possible improvements.

Projects evaluation

Projects will be evaluated with a score ranging from **0** to **8** points. Points will be added to the grade obtained in the **written exam** (**0** to **24**). In order to have your exam registered you have to pass both parts (obtain at least **4** points for the project and at least **15** for the written exam).

If project requirements are fulfilled you will have your project marked as passed.

You can either present your project before passing the written exam, or do the opposite.

Projects can be presented at any time (not related to exam dates).

Deadlines

Option 2 will be valid until **September 30th 2026**.

Project proposals will be accepted until **July 31st 2026**.

NB: if a student who has chosen Option 2 fails at presenting the project by September 30th, 2026, the assigned project will be considered void. In that case, the student must take the exam according to Option 1. If the student has already taken the written exam under Option 2 (shortened written exam), they will have to complete the assessment by taking the written exam to answer one question.

Possible ideas for projects

[DW] Select a set of data sources, integrate them by means of ETL operations (you can either use an integration tool or write simple scripts to solve common ETL problems), define the DFM model for your data, define and populate the corresponding star schema or snowflake schema, by using a relational DB (e.g., Postgres). Perform OLAP queries.

Focus: ability to combine data from different datasets in order to produce a relevant unified source of analysis

[DW] Identify a problem concerning data analysis and try to compare different approaches to address the proposed problem (e.g., carrying on the analysis via a relational DBMS and through a DW tool)

Focus: ability to compare different technologies

[DW] Select a dataset concerning a domain of your interest, carry out some relevant analyses and produce some report to show the results

Focus: ability to single out an interesting phenomenon (not explicitly evident from the data) and show how technology can help in communicating the findings of the analysis

[NoSQL] Select a dataset, a graph database tool, and a relational DBMS, and compare the results of analyses on different systems, highlighting advantages and disadvantages of each approach in terms of efficiency

Focus: ability to compare different technologies

[NoSQL] Select a graph database tool, acquire a reasonable knowledge on it and develop a simple DB project using such a tool.

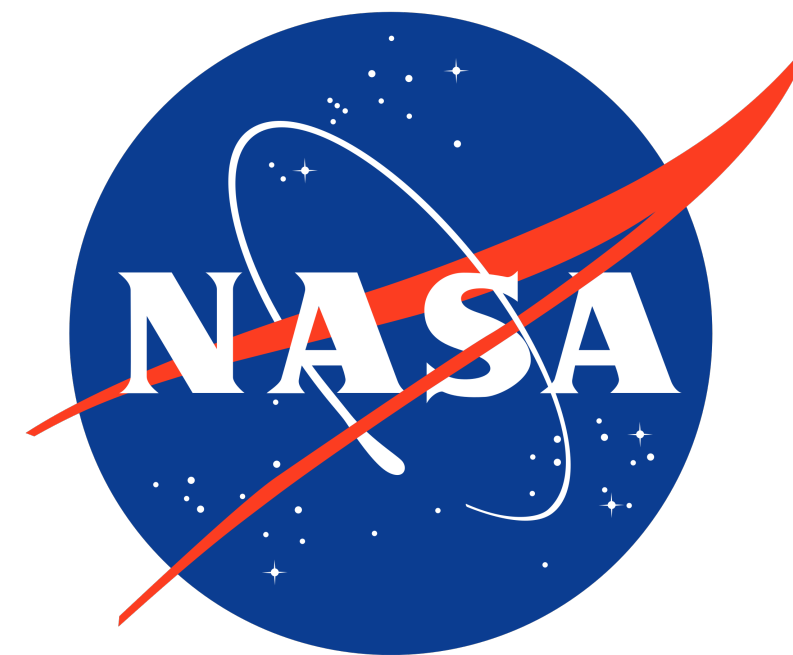
Focus: ability to acquire new knowledge about a DM tool and use it in a “real case” scenario)

Where to find datasets?

This is a non-exhaustive list of sources where you can find interesting datasets:

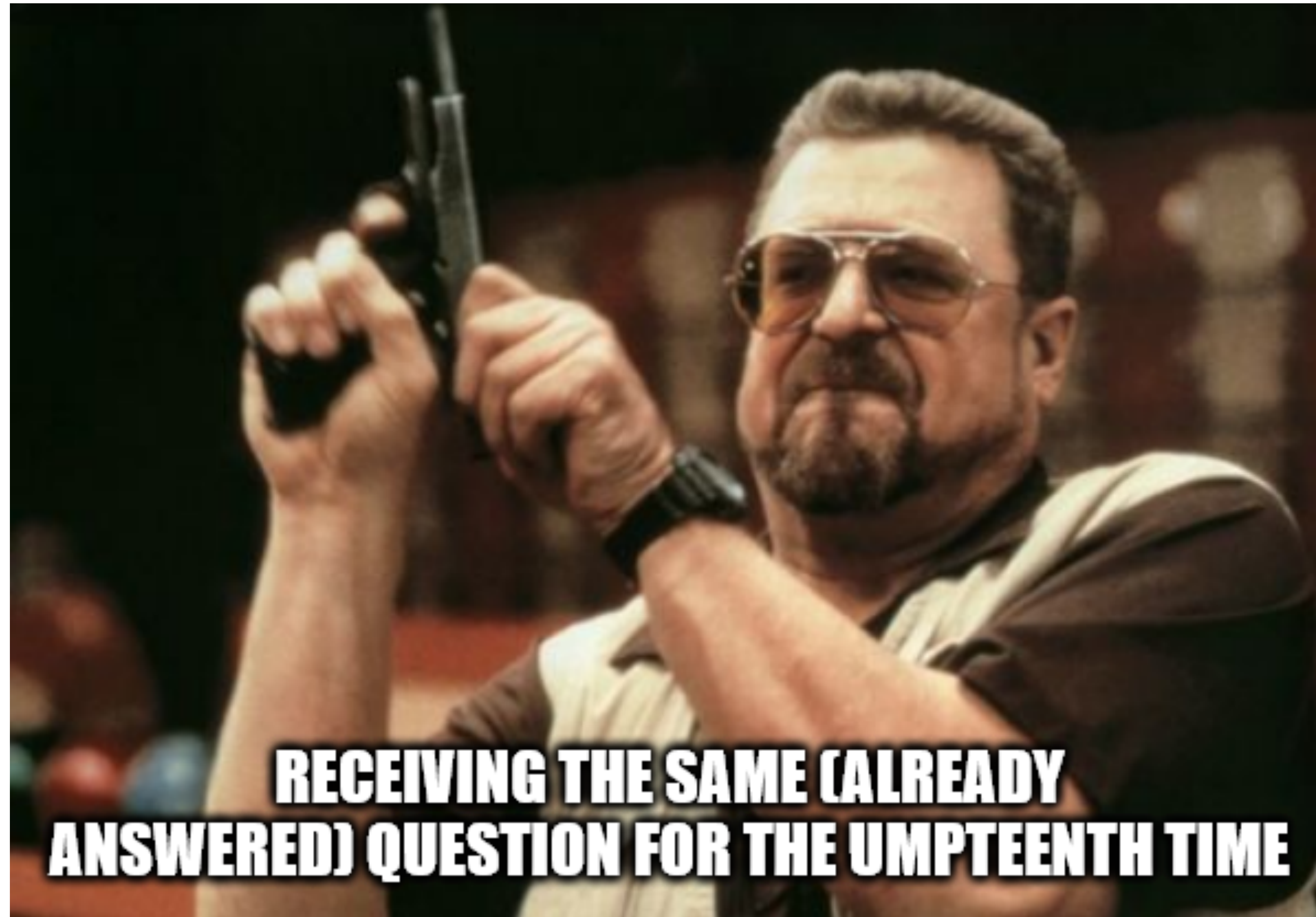
- kaggle.com/datasets
- data.europa.eu
- opendata.cern.ch
- earthdata.nasa.gov
- dati.gov.it (Italian only)
- data.gov
- datasetsearch.research.google.com

kaggle



data.europa.eu

Don't F.A.Q. with me!



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To avoid getting me upset, be sure to look at the multiple sources of information at your disposal before asking questions:

- Website of the course: <https://classroom.google.com/c/MTYyMzM4OTk1MTQx>
- My website (F.A.Q.): <https://sites.google.com/diag.uniroma1.it/rmdelfino/teaching/dm-2025-2026>
- These slides

Some advices

- Try to challenge yourselves, but do not underestimate the workload
- Don't let your partner do all the work and don't do all the work yourself: cooperate!
- Put your focus on the content, not on the tool (I won't buy anything)
- Take it as an opportunity to improve your writing and your communication skills

